

Problem_Set_2

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Introduction

This assignment focuses on the application of linear regression techniques within a predictive analytics framework. The objective is to estimate model parameters, assess model assumptions, and interpret results using both numerical summaries and visual diagnostics. All analyses are conducted using R to ensure reproducibility.

```
rm(list = ls())
```

1. Problem to demonstrate that the population regression line is fixed, but least square regression line varies

Suppose the population regression line is given by $Y = 2 + 3x$, while the data comes from the model $y = 2 + 3x + \varepsilon$.

Step 1: For x in the range $[5, 10]$ graph the population regression line.

Step 2: Generate $x_i (i = 1, 2, \dots, n)$ from $\text{Uniform}(5, 10)$ and $\varepsilon_i (i = 1, 2, \dots, n)$ from $N(0, 4^2)$. Hence, compute $y_1, y_2, \dots, y_n$.

Step 3: On the basis of the data $(x_i, y_i) (i = 1, 2, \dots, n)$ generated in Step 2, report the least squares regression line.

Step 4: Repeat steps 2-3 five times. Graph the 5 least squares regression lines over the population regression line obtained in Step 1. Interpret the findings.

Take $n = 50$. Set the seed as $\text{seed} = 123$.

```
set.seed(123)
x1=seq(5,10,length.out=200);x1

## [1] 5.000000 5.025126 5.050251 5.075377 5.100503 5.125628
5.150754
## [8] 5.175879 5.201005 5.226131 5.251256 5.276382 5.301508
5.326633
## [15] 5.351759 5.376884 5.402010 5.427136 5.452261 5.477387
5.502513
## [22] 5.527638 5.552764 5.577889 5.603015 5.628141 5.653266
5.678392
## [29] 5.703518 5.728643 5.753769 5.778894 5.804020 5.829146
5.854271
## [36] 5.879397 5.904523 5.929648 5.954774 5.979899 6.005025
6.030151
```

```
## [43] 6.055276 6.080402 6.105528 6.130653 6.155779 6.180905
6.206030
## [50] 6.231156 6.256281 6.281407 6.306533 6.331658 6.356784
6.381910
## [57] 6.407035 6.432161 6.457286 6.482412 6.507538 6.532663
6.557789
## [64] 6.582915 6.608040 6.633166 6.658291 6.683417 6.708543
6.733668
## [71] 6.758794 6.783920 6.809045 6.834171 6.859296 6.884422
6.909548
## [78] 6.934673 6.959799 6.984925 7.010050 7.035176 7.060302
7.085427
## [85] 7.110553 7.135678 7.160804 7.185930 7.211055 7.236181
7.261307
## [92] 7.286432 7.311558 7.336683 7.361809 7.386935 7.412060
7.437186
## [99] 7.462312 7.487437 7.512563 7.537688 7.562814 7.587940
7.613065
## [106] 7.638191 7.663317 7.688442 7.713568 7.738693 7.763819
7.788945
## [113] 7.814070 7.839196 7.864322 7.889447 7.914573 7.939698
7.964824
## [120] 7.989950 8.015075 8.040201 8.065327 8.090452 8.115578
8.140704
## [127] 8.165829 8.190955 8.216080 8.241206 8.266332 8.291457
8.316583
## [134] 8.341709 8.366834 8.391960 8.417085 8.442211 8.467337
8.492462
## [141] 8.517588 8.542714 8.567839 8.592965 8.618090 8.643216
8.668342
## [148] 8.693467 8.718593 8.743719 8.768844 8.793970 8.819095
8.844221
## [155] 8.869347 8.894472 8.919598 8.944724 8.969849 8.994975
9.020101
## [162] 9.045226 9.070352 9.095477 9.120603 9.145729 9.170854
9.195980
## [169] 9.221106 9.246231 9.271357 9.296482 9.321608 9.346734
9.371859
## [176] 9.396985 9.422111 9.447236 9.472362 9.497487 9.522613
9.547739
## [183] 9.572864 9.597990 9.623116 9.648241 9.673367 9.698492
9.723618
## [190] 9.748744 9.773869 9.798995 9.824121 9.849246 9.874372
9.899497
## [197] 9.924623 9.949749 9.974874 10.000000
```

```
y1=2+3*x1
```

```
plot(x1,y1,type='l',col="pink",lwd=3)
```

```
simulation_results <- data.frame(
```

```

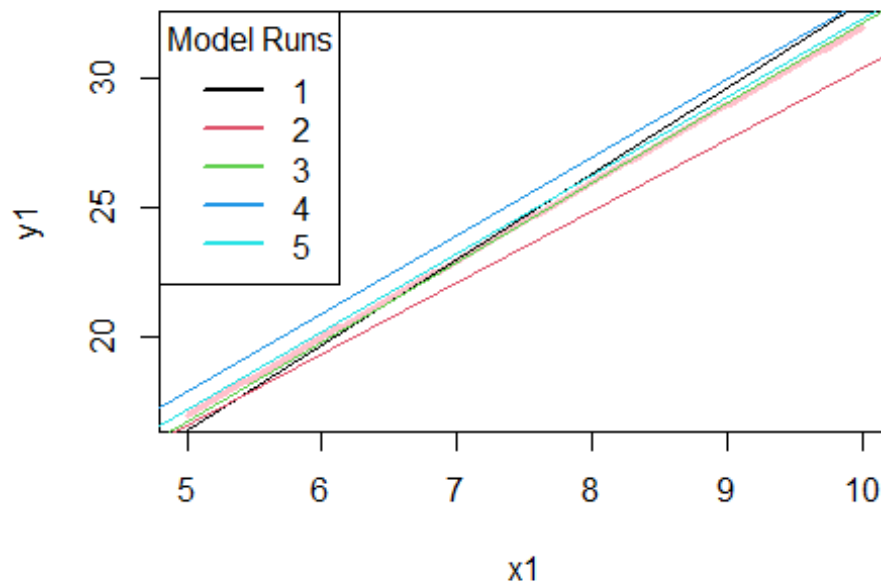
iteration = 1:5,
intercept = numeric(5),
slope = numeric(5)
)

for (i in 1:5){
  x = runif(50, 5, 10)
  eps = rnorm(50, 0, 4)
  y = 2 + 3*x + eps
  mod = lm(y ~ x)
  abline(mod, col = i)
  simulation_results$intercept[i] = coef(mod)[1]
  simulation_results$slope[i] = coef(mod)[2]
  print(summary(mod)$coefficients) # More detailed output
}

##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) -0.09638929  2.8260956 -0.03410688 9.729334e-01
## x           3.30539569  0.3651861  9.05126412 5.961769e-12
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  2.792188   3.7483426  0.7449128 4.599563e-01
## x           2.761042   0.4898233  5.6368122 8.950156e-07
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  1.392997   2.7885513  0.4995416 6.196801e-01
## x           3.073267   0.3670719  8.3723847 6.062269e-11
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  2.823089   2.7354376  1.032043 3.072243e-01
## x           3.023608   0.3623615  8.344175 6.681863e-11
##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  2.032506   3.1772492  0.6397063 5.254052e-01
## x           3.028097   0.4091446  7.4010440 1.789935e-09

legend("topleft",
      legend = paste(1:5),
      col = 1:5,
      lty = 1,
      lwd = 2,
      title = "Model Runs")

```



```
simulation_results
```

```
## iteration intercept slope
## 1 1 -0.09638929 3.305396
## 2 2 2.79218839 2.761042
## 3 3 1.39299737 3.073267
## 4 4 2.82308856 3.023608
## 5 5 2.03250638 3.028097
```

2. Problem to demonstrate that $\hat{\beta}_0$ and $\hat{\beta}$ minimizes RSS

Step 1: Generate x_i from Uniform(5, 10) and mean center the values. Generate ϵ_i from $N(0, 1)$. Calculate $y_i = 2 + 3x_i + \epsilon_i$, $i = 1, 2, \dots, n$. Take $n=50$ and seed=123.

Step 2: Now imagine that you only have the data on (x_i, y_i) , $i = 1, 2, \dots, n$, without knowing the mechanism that was used to generate the data in step 1. Assuming a linear regression of the type $y_i = \beta_0 + \beta x_i + \epsilon_i$, and based on these data (x_i, y_i) , $i = 1, 2, \dots, n$, obtain the least squares estimates of β_0 and β .

Step 3: Take a large number of grid values of (β_0, β) that also include the least squares estimates obtained from step 2. Compute the RSS for each parametric choice of (β_0, β) , where $RSS = (y_1 - \beta_0 - \beta x_1)^2 + (y_2 - \beta_0 - \beta x_2)^2 + \dots + (y_n - \beta_0 - \beta x_n)^2$. Find out for which combination of (β_0, β) , RSS is minimum.

```
n <- 50
xi <- runif(n, 5, 10)
xi_std <- xi - mean(xi)
```

```

eps <- rnorm(n, 0, 1)
y <- 2 + 3 * xi_std + eps

fitting <- lm(y ~ xi_std)
beta_not_hat <- coef(fitting)[1]
beta_hat <- coef(fitting)[2]

beta_not_grid <- seq(beta_not_hat - 2, beta_not_hat + 2, length = 51)
beta_grid <- seq(beta_hat - 2, beta_hat + 2, length = 51)

RSS <- matrix(NA, 50, 50)

for (i in 1:50) {
  for (j in 1:50) {
    RSS[i, j] <- sum((y - beta_not_grid[i] - beta_grid[j] * xi)^2)
  }
}
RSS

##           [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]
## [,8]
## [1,] 2335.716 2684.182 3067.398 3485.366 3938.085 4425.556 4947.777
##      5504.750
## [2,] 2385.694 2738.797 3126.651 3549.256 4006.613 4498.721 5025.580
##      5587.190
## [3,] 2436.312 2794.052 3186.544 3613.786 4075.780 4572.526 5104.022
##      5670.270
## [4,] 2487.570 2849.948 3247.077 3678.957 4145.588 4646.970 5183.104
##      5753.989
## [5,] 2539.468 2906.483 3308.249 3744.767 4216.036 4722.055 5262.827
##      5838.349
## [6,] 2592.006 2963.659 3370.062 3811.217 4287.123 4797.780 5343.189
##      5923.349
## [7,] 2645.184 3021.474 3432.515 3878.307 4358.851 4874.145 5424.191
##      6008.989
## [8,] 2699.002 3079.929 3495.608 3946.037 4431.218 4951.150 5505.834
##      6095.268
## [9,] 2753.460 3139.025 3559.341 4014.408 4504.226 5028.795 5588.116
##      6182.188
## [10,] 2808.558 3198.760 3623.713 4083.418 4577.873 5107.080 5671.038
##      6269.748
## [11,] 2864.296 3259.136 3688.726 4153.068 4652.161 5186.005 5754.601
##      6357.948
## [12,] 2920.675 3320.151 3754.379 4223.358 4727.089 5265.570 5838.803
##      6446.787
## [13,] 2977.693 3381.807 3820.672 4294.288 4802.656 5345.775 5923.645
##      6536.267
## [14,] 3035.351 3444.102 3887.605 4365.859 4878.864 5426.620 6009.128
##      6626.387

```

[15,] 3093.649 3507.037 3955.177 4438.069 4955.711 5508.105 6095.250
6717.147
[16,] 3152.587 3570.613 4023.390 4510.919 5033.199 5590.230 6182.013
6808.546
[17,] 3212.165 3634.828 4092.243 4584.409 5111.326 5672.995 6269.415
6900.586
[18,] 3272.383 3699.684 4161.736 4658.539 5190.094 5756.400 6357.457
6993.266
[19,] 3333.241 3765.179 4231.869 4733.310 5269.502 5840.445 6446.140
7086.585
[20,] 3394.739 3831.314 4302.641 4808.720 5349.549 5925.130 6535.462
7180.545
[21,] 3456.877 3898.090 4374.054 4884.770 5430.237 6010.455 6625.424
7275.145
[22,] 3519.655 3965.505 4446.107 4961.460 5511.564 6096.420 6716.027
7370.385
[23,] 3583.073 4033.561 4518.800 5038.790 5593.532 6183.025 6807.269
7466.264
[24,] 3647.131 4102.256 4592.133 5116.760 5676.140 6270.270 6899.151
7562.784
[25,] 3711.829 4171.592 4666.105 5195.371 5759.387 6358.155 6991.674
7659.944
[26,] 3777.167 4241.567 4740.718 5274.621 5843.275 6446.680 7084.836
7757.744
[27,] 3843.145 4312.182 4815.971 5354.511 5927.802 6535.845 7178.638
7856.183
[28,] 3909.763 4383.438 4891.864 5435.041 6012.970 6625.650 7273.081
7955.263
[29,] 3977.021 4455.333 4968.397 5516.211 6098.777 6716.095 7368.163
8054.983
[30,] 4044.919 4527.869 5045.570 5598.022 6185.225 6807.180 7463.886
8155.343
[31,] 4113.457 4601.044 5123.382 5680.472 6272.313 6898.905 7560.248
8256.342
[32,] 4182.635 4674.860 5201.835 5763.562 6360.040 6991.270 7657.250
8357.982
[33,] 4252.453 4749.315 5280.928 5847.292 6448.408 7084.275 7754.893
8460.262
[34,] 4322.911 4824.410 5360.661 5931.662 6537.415 7177.920 7853.175
8563.182
[35,] 4394.009 4900.146 5441.034 6016.673 6627.063 7272.204 7952.097
8666.741
[36,] 4465.747 4976.521 5522.046 6102.323 6717.350 7367.129 8051.660
8770.941
[37,] 4538.125 5053.537 5603.699 6188.613 6808.278 7462.694 8151.862
8875.781
[38,] 4611.143 5131.192 5685.992 6275.543 6899.846 7558.899 8252.704
8981.261
[39,] 4684.801 5209.487 5768.925 6363.113 6992.053 7655.744 8354.187
9087.380

```

## [40,] 4759.099 5288.423 5852.498 6451.324 7084.901 7753.229 8456.309
9194.140
## [41,] 4834.037 5367.998 5936.710 6540.174 7178.388 7851.354 8559.071
9301.540
## [42,] 4909.615 5448.214 6021.563 6629.664 7272.516 7950.119 8662.474
9409.580
## [43,] 4985.833 5529.069 6107.056 6719.794 7367.284 8049.524 8766.516
9518.259
## [44,] 5062.692 5610.565 6193.189 6810.564 7462.691 8149.569 8871.198
9627.579
## [45,] 5140.190 5692.700 6279.962 6901.975 7558.739 8250.254 8976.521
9737.539
## [46,] 5218.328 5775.475 6367.374 6994.025 7655.426 8351.579 9082.483
9848.139
## [47,] 5297.106 5858.891 6455.427 7086.715 7752.754 8453.544 9189.086
9959.378
## [48,] 5376.524 5942.946 6544.120 7180.045 7850.721 8556.149 9296.328
10071.258
## [49,] 5456.582 6027.642 6633.453 7274.015 7949.329 8659.394 9404.210
10183.778
## [50,] 5537.280 6112.977 6723.426 7368.625 8048.577 8763.279 9512.733
10296.937
##           [,9]      [,10]      [,11]      [,12]      [,13]      [,14]      [,15]
## [1,] 6096.474 6722.950 7384.176 8080.154 8810.884 9576.364 10376.60
## [2,] 6183.551 6814.664 7480.528 8181.144 8916.510 9686.628 10491.50
## [3,] 6271.269 6907.019 7577.520 8282.773 9022.777 9797.532 10607.04
## [4,] 6359.626 7000.013 7675.152 8385.042 9129.684 9909.076 10723.22
## [5,] 6448.623 7093.648 7773.424 8487.952 9237.230 10021.260 10840.04
## [6,] 6538.260 7187.922 7872.336 8591.501 9345.417 10134.084 10957.50
## [7,] 6628.537 7282.837 7971.888 8695.690 9454.244 10247.548 11075.60
## [8,] 6719.454 7378.391 8072.080 8800.519 9563.710 10361.652 11194.35
## [9,] 6811.011 7474.586 8172.912 8905.989 9673.817 10476.397 11313.73
## [10,] 6903.208 7571.420 8274.384 9012.098 9784.564 10591.781 11433.75
## [11,] 6996.046 7668.895 8376.495 9118.847 9895.950 10707.805 11554.41
## [12,] 7089.523 7767.009 8479.247 9226.237 10007.977 10824.469 11675.71
## [13,] 7183.640 7865.764 8582.639 9334.266 10120.644 10941.773 11797.65
## [14,] 7278.397 7965.158 8686.671 9442.935 10233.950 11059.717 11920.23
## [15,] 7373.794 8065.193 8791.343 9552.244 10347.897 11178.301 12043.46
## [16,] 7469.831 8165.867 8896.655 9662.194 10462.484 11297.525 12167.32
## [17,] 7566.508 8267.182 9002.607 9772.783 10577.710 11417.389 12291.82
## [18,] 7663.826 8369.137 9109.199 9884.012 10693.577 11537.893 12416.96
## [19,] 7761.783 8471.731 9216.431 9995.882 10810.084 11659.037 12542.74
## [20,] 7860.380 8574.966 9324.303 10108.391 10927.230 11780.821 12669.16
## [21,] 7959.617 8678.840 9432.814 10221.540 11045.017 11903.245 12796.22
## [22,] 8059.494 8783.355 9541.966 10335.329 11163.444 12026.309 12923.93
## [23,] 8160.011 8888.509 9651.758 10449.759 11282.510 12150.013 13052.27
## [24,] 8261.168 8994.304 9762.190 10564.828 11402.217 12274.358 13181.25
## [25,] 8362.965 9100.738 9873.262 10680.537 11522.564 12399.342 13310.87
## [26,] 8465.403 9207.813 9984.974 10796.887 11643.551 12524.966 13441.13
## [27,] 8568.480 9315.527 10097.326 10913.876 11765.177 12651.230 13572.03

```

```

## [28,] 8672.197 9423.882 10210.318 11031.505 11887.444 12778.134 13703.57
## [29,] 8776.554 9532.876 10323.950 11149.775 12010.351 12905.678 13835.76
## [30,] 8881.551 9642.511 10438.222 11268.684 12133.897 13033.862 13968.58
## [31,] 8987.188 9752.785 10553.134 11388.233 12258.084 13162.686 14102.04
## [32,] 9093.465 9863.700 10668.685 11508.422 12382.911 13292.150 14236.14
## [33,] 9200.382 9975.254 10784.877 11629.252 12508.377 13422.254 14370.88
## [34,] 9307.940 10087.449 10901.709 11750.721 12634.484 13552.998 14506.26
## [35,] 9416.137 10200.283 11019.181 11872.830 12761.231 13684.382 14642.29
## [36,] 9524.974 10313.758 11137.293 11995.580 12888.617 13816.406 14778.95
## [37,] 9634.451 10427.872 11256.045 12118.969 13016.644 13949.070 14916.25
## [38,] 9744.568 10542.627 11375.437 12242.998 13145.311 14082.374 15054.19
## [39,] 9855.325 10658.021 11495.469 12367.667 13274.617 14216.319 15192.77
## [40,] 9966.722 10774.056 11616.141 12492.977 13404.564 14350.903 15331.99
## [41,] 10078.759 10890.730 11737.453 12618.926 13535.151 14486.127 15471.85
## [42,] 10191.437 11008.045 11859.404 12745.515 13666.377 14621.991 15612.36
## [43,] 10304.754 11125.999 11981.996 12872.745 13798.244 14758.495 15753.50
## [44,] 10418.711 11244.594 12105.228 13000.614 13930.751 14895.639 15895.28
## [45,] 10533.308 11363.828 12229.100 13129.123 14063.897 15033.423 16037.70
## [46,] 10648.545 11483.703 12353.612 13258.272 14197.684 15171.847 16180.76
## [47,] 10764.422 11604.217 12478.764 13388.062 14332.111 15310.911 16324.46
## [48,] 10880.939 11725.372 12604.556 13518.491 14467.177 15450.615 16468.80
## [49,] 10998.097 11847.167 12730.988 13649.560 14602.884 15590.959 16613.79
## [50,] 11115.894 11969.601 12858.060 13781.270 14739.231 15731.943 16759.41
##      [,16]  [,17]  [,18]  [,19]  [,20]  [,21]  [,22]
##      [,23]
## [1,] 11211.58 12081.31 12985.80 13925.04 14899.02 15907.76 16951.25
18029.49
## [2,] 11331.12 12205.49 13114.61 14058.49 15037.11 16050.49 17098.62
18181.50
## [3,] 11451.30 12330.31 13244.07 14192.58 15175.84 16193.85 17246.62
18334.14
## [4,] 11572.12 12455.76 13374.16 14327.31 15315.21 16337.86 17395.26
18487.42
## [5,] 11693.57 12581.86 13504.89 14462.68 15455.22 16482.51 17544.55
18641.34
## [6,] 11815.67 12708.59 13636.27 14598.69 15595.87 16627.79 17694.47
18795.90
## [7,] 11938.41 12835.97 13768.28 14735.34 15737.15 16773.72 17845.03
18951.10
## [8,] 12061.79 12963.99 13900.93 14872.63 15879.08 16920.28 17996.24
19106.94
## [9,] 12185.81 13092.64 14034.23 15010.56 16021.65 17067.49 18148.08
19263.42
## [10,] 12310.47 13221.94 14168.16 15149.13 16164.86 17215.33 18300.56
19420.54
## [11,] 12435.77 13351.88 14302.73 15288.35 16308.71 17363.82 18453.68
19578.30
## [12,] 12561.71 13482.45 14437.95 15428.20 16453.20 17512.95 18607.45
19736.70
## [13,] 12688.28 13613.67 14573.80 15568.69 16598.32 17662.71 18761.85

```


19895.74
[14,] 12815.50 13745.52 14710.30 15709.82 16744.09 17813.12 18916.89
20055.42
[15,] 12943.36 13878.02 14847.43 15851.59 16890.50 17964.16 19072.58
20215.74
[16,] 13071.86 14011.16 14985.20 15994.00 17037.55 18115.85 19228.90
20376.70
[17,] 13201.00 14144.93 15123.62 16137.05 17185.24 18268.17 19385.86
20538.30
[18,] 13330.78 14279.35 15262.67 16280.74 17333.57 18421.14 19543.47
20700.54
[19,] 13461.20 14414.41 15402.36 16425.07 17482.53 18574.75 19701.71
20863.42
[20,] 13592.26 14550.10 15542.70 16570.04 17632.14 18728.99 19860.59
21026.95
[21,] 13723.96 14686.44 15683.67 16715.66 17782.39 18883.88 20020.12
21191.11
[22,] 13856.29 14823.41 15825.28 16861.91 17933.28 19039.40 20180.28
21355.91
[23,] 13989.27 14961.03 15967.54 17008.80 18084.81 19195.57 20341.08
21521.35
[24,] 14122.89 15099.29 16110.43 17156.33 18236.98 19352.38 20502.53
21687.43
[25,] 14257.15 15238.18 16253.97 17304.50 18389.78 19509.82 20664.61
21854.15
[26,] 14392.05 15377.72 16398.14 17453.31 18543.23 19667.91 20827.33
22021.51
[27,] 14527.59 15517.89 16542.95 17602.76 18697.32 19826.63 20990.70
22189.51
[28,] 14663.77 15658.71 16688.41 17752.85 18852.05 19986.00 21154.70
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## [1,]    1   1
```

3. Problem to demonstrate that least square estimators are unbiased

Step 1: Generate x_i ($i = 1, 2, \dots, n$) from Uniform(0, 1), ε_i ($i = 1, 2, \dots, n$) from $N(0, 1)$ and hence generate y using $y_i = \beta_0 + \beta x_i + \varepsilon_i$. (Take $\beta_0 = 2$, $\beta = 3$).

Step 2: On the basis of the data (x_i, y_i) ($i = 1, 2, \dots, n$) generated in Step 1, obtain the least square estimates of β_0 and β .

Repeat Steps 1-2, $R = 1000$ times. In each simulation obtain $\hat{\beta}_0$ and $\hat{\beta}$. Finally, the least-square estimates will be given by the average of these estimated values. Compare these with the true β_0 and β and comment.

Take $n = 50$ and $\text{seed} = 123$.

```
set.seed(123)
n3 <- 50
R <- 10000

beta_not_hat_vec <- array()
beta_hat_vec <- array()
eps3 <- rnorm(n3,0,1)

for(r in 1:R){
  x3 <- runif(n3,0,1)
  y3 <- 2 + 3*x3 + eps3

  fit3 <- lm(y3 ~ x3)
  beta_not_hat_vec[r] <- coef(fit3)[1]
  beta_hat_vec[r] <- coef(fit3)[2]
}

predicted_beta_not_hat <- sum(beta_not_hat_vec)/R
predicted_beta_not_hat

## [1] 2.033112

predicted_beta_hat_vec <- sum(beta_hat_vec)/R
predicted_beta_hat_vec

## [1] 3.00193
```

4. Comparing several simple linear regressions

Attach “Boston” data from MASS library in R. Select median value of owner-occupied homes, as the response and per capita crime rate, nitrogen oxides concentration, proportion of blacks and percentage of lower status of the population as predictors.

- Selecting the predictors one by one, run four separate linear regressions to the data. Present the output in a single table.
- Which model gives the best fit?
- Compare the coefficients of the predictors from each model and comment on the usefulness of the predictors.

```
library(MASS)
data(Boston)

fit1 <- lm(medv ~ crim, data=Boston)
fit2 <- lm(medv ~ nox, data=Boston)
fit3 <- lm(medv ~ black, data=Boston)
fit4 <- lm(medv ~ lstat, data=Boston)

summary_t <- data.frame(
```

```

Model = c("crim", "nox", "black", "lstat"),
Intercept = c(coef(fit1)[1], coef(fit2)[1], coef(fit3)[1], coef(fit4)[1]),
Slope = c(coef(fit1)[2], coef(fit2)[2], coef(fit3)[2], coef(fit4)[2]),
R2 = c(summary(fit1)$r.squared,
        summary(fit2)$r.squared,
        summary(fit3)$r.squared,
        summary(fit4)$r.squared)
)

print(summary_t)

##      Model Intercept      Slope      R2
## crim    crim  24.03311 -0.41519028 0.1507805
## nox     nox   41.34587 -33.91605501 0.1826030
## black  black  10.55103  0.03359306 0.1111961
## lstat  lstat  34.55384 -0.95004935 0.5441463

best_model <- summary_t[which.max(summary_t$R2), ]
best_model

##      Model Intercept      Slope      R2
## lstat  lstat  34.55384 -0.9500494 0.5441463

```